

Generating sequences and finding the nth term

From “add four each time” to a rule that reaches the hundredth term at once.
[Cambridge insert]

LESSON 137–138

2 × 40 MIN

MATEMATIKA 6, QO‘SHIMCHA MAVZU

S7 9 SEQUENCES AND FUNCTIONS

LESSON CLOCK

20'

20'

25'

10'

5'

Term-to-term rules	20 min
From position to term	20 min
Finding the nth term	25 min
Using the rule	10 min
Homework	5 min

LEARNING OBJECTIVES

- Continue a sequence from its term-to-term rule.
- Generate terms from a position-to-term rule such as $3n + 2$.
- Find the nth term of a linear sequence from its common difference.
- Decide whether a given number is a term, and say which one.

TEXTBOOK

National	pp. 395–399
Cambridge	Stage 7, pp. 90–97
Workbook	Exercise 9.1–9.3

01

Explanation

Term-to-term rules

A sequence is a list of numbers in a fixed order. Each number is a **term**, and its **position** is $n = 1, 2, 3, \dots$. The easiest rule says what to do to one term to get the next.

SEQUENCE	TERM-TO-TERM RULE	NEXT TWO TERMS
3, 7, 11, 15, ...	add 4	19, 23
5, 8, 11, 14, ...	add 3	17, 20
20, 17, 14, 11, ...	subtract 3	8, 5
2, 4, 8, 16, ...	multiply by 2	32, 64

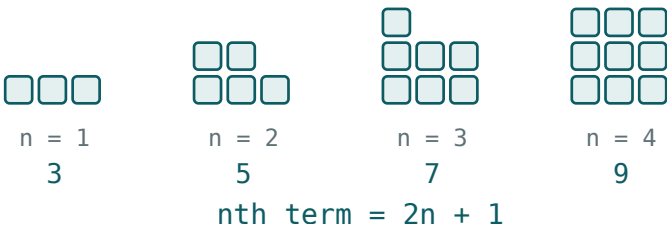
A TERM-TO-TERM RULE CANNOT JUMP

To reach the 100th term of the first sequence this way you would add 4 ninety-nine times. That is why the next rule is worth having.

From position to term

A **position-to-term** rule — the n th term — takes the position straight to the term.
Substitute $n = 1, 2, 3, \dots$ and the sequence appears.

NTH TERM	$N = 1$	$N = 2$	$N = 3$	$N = 4$
$3n + 2$	5	8	11	14
$4n - 1$	3	7	11	15
$23 - 3n$	20	17	14	11
n^2	1	4	9	16



A pattern of squares — the count rises by two each time, so the n th term is $2n + 1$

THE RULE REACHES ANY TERM IN ONE STEP

The 100th term of $4n - 1$ is $4 \cdot 100 - 1 = 399$. No list, no adding ninety-nine times.

Finding the nth term

A sequence that goes up (or down) by the same amount every time is **linear**. Call that amount the common difference d , and the first term a .

$n\text{th term} = dn + (a - d)$

The dn builds the steady climb; the bracket is the adjustment that makes the first term come out right. In practice: write dn , work out the first term it gives, and correct it.

SEQUENCE	D	DN GIVES AT $N = 1$	CORRECTION	NTH TERM
5, 8, 11, 14	3	3	+2	$3n + 2$
2, 9, 16, 23	7	7	-5	$7n - 5$
6, 10, 14, 18	4	4	+2	$4n + 2$
20, 17, 14, 11	-3	-3	+23	$23 - 3n$

CHECK THE RULE ON THE FIRST TWO TERMS

A rule that gives the right first term but the wrong second one has the wrong d ; one that climbs correctly but starts wrong needs a different correction. Both take five seconds to test.

Using the rule

Once the n th term is known, two questions can be answered that a list cannot answer at all.

QUESTION	WORKING	ANSWER
the 50th term of $3n + 2$	$3 \cdot 50 + 2$	152
is 47 a term of $3n + 2$?	$3n = 45, n = 15$	yes, the 15th
is 60 a term of $4n - 1$?	$4n = 61$	no — n is not whole
the first negative term of $23 - 3n$	$23 - 3n < 0, n > 7.67$	$n = 8$, term -1

THE POSITION MUST BE A WHOLE NUMBER

That is the whole test for membership. Solve for n ; if it comes out a whole number the value is a term, and n says which one.

02 Worked examples

EXAMPLE 1 Find the n th term of 5, 8, 11, 14, ... and its 20th term.

- ① Common difference $d = 3$, so start from $3n$.
- ② $3n$ gives 3 at $n = 1$, but the term is 5: add 2.
- ③ n th term $= 3n + 2$; at $n = 20$, $3 \cdot 20 + 2 = 62$.
Check $n = 2$: 8 ✓

ANSWER

$3n + 2$, and 62

EXAMPLE 2 Find the n th term of 20, 17, 14, 11, ... and its first negative term.

- ① $d = -3$, so start from $-3n$.
- ② $-3n$ gives -3 at $n = 1$, but the term is 20: add 23.
- ③ n th term $= 23 - 3n$. It is negative when $n > 7\frac{2}{3}$, so $n = 8$ gives -1 .

ANSWER

$23 - 3n$; the 8th term, -1

EXAMPLE 3 Is 47 a term of the sequence with n th term $3n + 2$?

- ① Set $3n + 2 = 47$.
- ② $3n = 45$.
- ③ $n = 15$ — a whole number, so yes.
Check: $3 \cdot 15 + 2 = 47$ ✓

ANSWER

Yes, the 15th term

03 Interactive model

Ask for the 100th term before giving the n th-term rule; the impatience the class feels is exactly the reason the rule exists.

Position, term and rule

In 3, 7, 11, 15 the common difference is:

The n th term of 5, 8, 11, 14 is:

The 10th term of $4n - 1$ is:

The n th term of 20, 17, 14 is:

Is 60 a term of $4n - 1$?

1, 4, 9, 16 is generated by:

A term-to-term rule of “multiply by 2” makes the sequence:

The n th term of 6, 10, 14, 18 is:

$$4n + 2$$

$$4n$$

$$6n$$

$$2n + 4$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Sequence	Ketma-ketlik	Последовательность
Term	Had	Член последовательности
Position	O‘rin (tartib raqami)	Номер (позиция)
Term-to-term rule	Haddan hadga o‘tish qoidasi	Рекуррентное правило
n th term	n -chi had	n -й член
Common difference	Ayirma	Разность
Linear sequence	Chiziqli ketma-ketlik	Линейная последовательность
Square numbers	Kvadrat sonlar	Квадратные числа

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The common difference of 2, 9, 16, 23 is:

The n th term of 2, 9, 16, 23 is:

A position-to-term rule lets you:

The 100th term of $4n - 1$ is:

A number is a term only when n comes out:

In $dn + (a - d)$ the letter d is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The next two terms of 3, 7, 11, 15

ANSWER 19, 23

2. The next two terms of 20, 17, 14

ANSWER 11, 8

3. The first three terms of $5n$

ANSWER 5, 10, 15

4. The first three terms of $2n + 3$

ANSWER 5, 7, 9

5. The 10th term of $4n - 1$

ANSWER 39

6. The term-to-term rule of 5, 8, 11

ANSWER Add 3

7. The first term of $3n + 2$

ANSWER 5

Medium · the standard question

7 PROBLEMS

1. The n th term of 5, 8, 11, 14

ANSWER $3n + 2$

2. The n th term of 2, 9, 16, 23

ANSWER $7n - 5$

3. The n th term of 6, 10, 14, 18

ANSWER $4n + 2$

4. The n th term of 20, 17, 14, 11

ANSWER $23 - 3n$

5. The 100th term of $4n - 1$

ANSWER 399

6. The 50th term of $3n + 2$

ANSWER 152

7. Is 47 a term of $3n + 2$?

ANSWER Yes, the 15th

Hard · stretch and reason

7 PROBLEMS

1. Is 60 a term of $4n - 1$?

ANSWER No — $4n = 61$ is not whole

2. Which term of $7n - 5$ equals 100?

ANSWER The 15th

3. The n th term of 1, 4, 9, 16

ANSWER n^2

4. The n th term of 1, 3, 6, 10

ANSWER $n(n + 1) \div 2$

5. The first term of $4n + 2$ above 100

ANSWER 102, at $n = 25$

6. A sequence starts at 4 and adds 6: its n th term

ANSWER $6n - 2$

7. The first negative term of $23 - 3n$

ANSWER -1, at $n = 8$

07 Homework

Homework — 5 tasks

Test every n th term you write on the first two terms before moving on.

- Write the first five terms of the sequence with n th term $6n - 1$.
- Find the n th term of 7, 12, 17, 22,
- Find the n th term of 30, 26, 22, 18,
- Is 82 a term of the sequence with n th term $5n + 2$? Say which term, or why not.
- A pattern of squares uses 4, 7, 10, ... tiles. Find the n th term and the 50th figure.